TEXORA · LECTURE SHEET

Detailed Study of Modern Long Staple Spinning Systems

Long fibres need pinned control, patient preparation and — in the worsted system — combing. This sheet works through the worsted sequence stage by stage, sets semi-worsted and woollen spinning beside it, and closes with the modern developments that are changing gilling, combing and spinning.

YM-4201 · MODERN YARN MANUFACTURING · TOPIC 5 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. What “long staple” means in spinning

A long-staple system handles fibres whose length is too great for cotton roller drafting settings — wool for worsted and woollen yarns, long bast fibres, and long-cut synthetic staple made to imitate them. Three features define such systems wherever they occur. First, drafting is controlled by **pins** — gill fallers or comb pins — that enter the fibre fringe and carry short and medium fibres forward at a controlled speed, because rollers alone, set far apart for long fibres, would leave a large uncontrolled floating zone. Second, preparation is long: many doublings and repeated passages average out irregularity before spinning. Third, the systems divide by what they do about *short fibre* and *fibre parallelisation* — and that division gives the three classical systems of this sheet: worsted, semi-worsted and woollen.

2. Fibre requirements

Long-staple spinning begins with fibre selection, and each system demands a different fibre.

- **Length and length uniformity.** Worsted spinning needs long, uniform fibre: short fibres will be combed out as noil, so feeding short fibre to a worsted line is paying to manufacture waste. A narrow length distribution matters as much as mean length, because gill and roller settings are a compromise for the whole population.
- **Fineness.** Yarn count is limited by the number of fibres in the yarn cross-section. Fine worsted yarns require fine wool; no drafting can put fibres into a cross-section that the raw material does not supply.
- **Strength and cleanliness.** Vegetable matter, dust and second cuts raise carding and combing waste and cause yarn faults. Tender (weakened) wool breaks in carding and combing and reappears as

extra noil.

- **For synthetic long staple.** Man-made fibres are cut or tow-converted to the length the system requires, with crimp added to imitate wool's cohesion; their uniformity is their advantage, their static and melting behaviour their handling problems.

3. The worsted system, stage by stage

The worsted system produces smooth, strong, comparatively fine yarns from parallel fibres with short fibre removed. Its sequence is: scouring → drying → carding → gilling (pre-comb preparation) → combing → gilling (post-comb) → drawing → roving → spinning → winding and twisting.

3.1 Scouring

Raw (greasy) wool carries grease, suint, dirt and vegetable matter. **Scouring** washes it in successive bowls of warm water with detergent and alkali, followed by drying, to a controlled residual grease level: over-scoured wool is harsh and breaks; under-scoured wool mats and loads clothing. Lots are blended at this stage, while the fibre is loose.

3.2 Carding

The worsted card (a roller card with swift, workers, strippers and doffer, fed from a hopper and lattice) opens the locks, removes much of the vegetable matter and dirt, begins parallelisation and delivers a sliver. Worsted carding is gentler than woollen carding in intention: the object is an opened, webbed sliver with fibres already partly aligned — the raw material of a parallel top — not the deliberately randomised condenser feed of the woollen system. Bur removal and the condition of the card clothing decide how much vegetable fault survives to trouble the comb.

3.3 Gilling before combing

Card sliver is passed through **gill boxes** (intersecting gills): two banks of pinned faller bars, an upper and a lower set, whose pins intersect in the drafting zone. The fallers move the fibre forward at back-roller speed and step aside near the front rollers, so fibres are positively controlled almost to the nip that drafts them. Several gilling passages with doublings straighten fibres, hook out carding bends, and produce a sliver of the evenness and fibre alignment the comb requires: a comb fed with tangled sliver combs badly and wastes good fibre as noil.

3.4 Combing — the rectilinear comb

Combing is the defining worsted operation. The **rectilinear (French) comb** works in a cycle on successive fringes of sliver:

1. The sliver fringe is presented and gripped by a nipper.

2. A circular comb carrying fine pins combs through the protruding fringe, removing short fibres, neps and remaining vegetable matter into the noil.
3. Detaching rollers draw the combed fringe forward through a top comb, which combs the tail end of the fringe as it is detached.
4. Successive detached fringes overlap and are drawn together into a continuous combed sliver — the **top**.

The output divides into **top** (long, parallel, combed fibre) and **noil** (the short fibre and impurities removed). Noil is not a loss to hide: its percentage is the price of worsted quality, and it is sold for woollen and other uses. Too high a noil figure, however, usually signals short or tender raw material or poor pre-comb preparation rather than thorough combing.

> **WORKED EXAMPLE — COMBING NOIL PERCENTAGE**
>
> A combing set is fed with 250 kg of prepared sliver in a shift and produces 32.5 kg of noil.
> Noil % = 32.5 / 250 × 100 = **13.0%**.
> Top produced (ignoring other small waste) = 250 − 32.5 = **217.5 kg**.
> If the following shift, on the same raw material, returns 19% noil, the correct response is investigation — preparation, comb pins, settings, fibre tenderness — not acceptance: six extra kilograms in every hundred have moved from the valuable top into the low-value noil.

3.5 Gilling and drawing after combing

The top is gilled again — usually more than once — to restore evenness after the comb's intermittent action, then passes through a drawing sequence in which gill boxes and then roller-drafting draw frames progressively reduce sliver weight with doublings at each passage. The governing mass relation is the same at every passage:

$$\text{Output weight} = \text{feed weight} \times \text{doublings} / \text{draft}$$

> **WORKED EXAMPLE — DRAFT ACROSS A GILLING PASSAGE**
>
> A gill box is fed with 8 ends of sliver, each 12 g/m, and works with a draft of 10.
> Total feed weight = 8 × 12 = 96 g/m. Output weight = 96 / 10 = **9.6 g/m**.
> Equivalently: output = feed × doublings / draft = 12 × 8 / 10 = **9.6 g/m**.
> The passage has made the sliver finer (12 → 9.6 g/m in single-end terms) while eight-fold doubling has averaged the irregularities of the feed ends — draft and doubling working, as always, in opposite directions on weight and together on evenness.

3.6 Roving

The final drawn sliver is still far too heavy to spin directly to fine worsted counts. A **roving** stage gives it a high draft on a frame with apron or roller control suited to the now well-parallel fibres, inserting a small amount of twist — just enough to give the roving strength to unwind and feed at spinning without drafting out under its own tension. Roving twist is a handling aid, not yarn twist; excess roving twist resists drafting at the spinning frame and causes hard, uneven yarn.

3.7 Spinning

Worsted roving is spun principally on ring frames adapted to long staple: apron drafting, appropriate roller settings, and twist inserted by ring and traveller. The yarn is smooth, with fibres lying parallel along its axis — hence worsted fabric's clear weave definition, lustre, strength and comparatively low hairiness. Winding clears faults and builds the delivery package; two-fold twisting of single yarns is common in worsted practice because a plied structure balances twist liveliness and improves evenness and strength for weaving yarns.

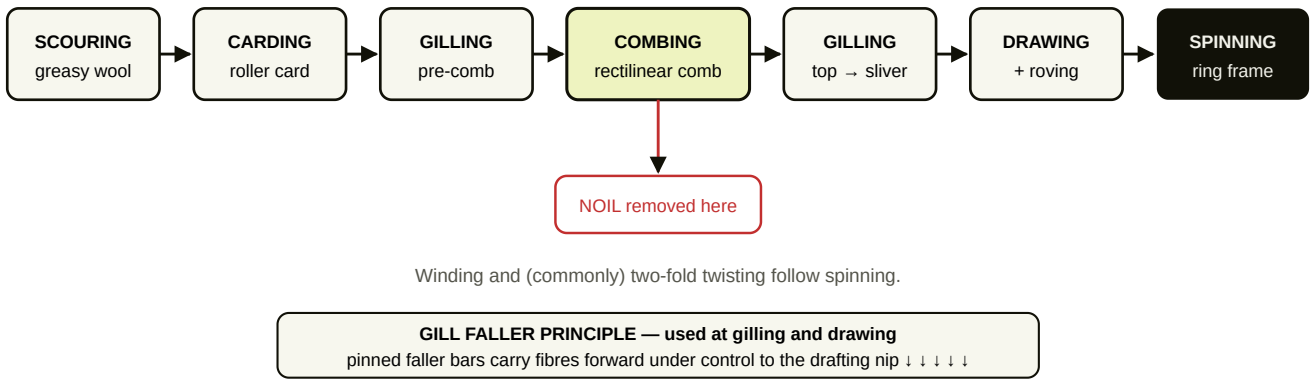


Figure 1 — The worsted processing line, with combing as the defining operation, and the gill faller principle that controls drafting in the preparation stages.

4. Semi-worsted and woollen systems compared

The **semi-worsted** system follows the worsted route — carding, gilling, drawing, roving, spinning — but **omits combing**. Fibres stay less perfectly parallel and short fibre stays in the yarn, so the yarn is bulkier, softer and cheaper than a true worsted yarn, and coarser in its practical count range. It is widely used for knitting yarns, blankets and furnishing yarns, and for long-cut synthetics, where the raw material is uniform enough not to need combing. The **woollen** system is different in aim, not merely shorter: fibre — often shorter wool, noil and recovered material, blended and oiled — is carded on a woollen card set ending in a **condenser**, which divides the web into strips and rubs them into rovings (**slubbings**) with fibres deliberately left in a disordered, lofty arrangement, then spun, traditionally on a mule and today also on ring and other frames. There is no gilling, drawing or combing: evenness comes

from blending, carding and condensation, and the yarn's character — soft, bulky, warm, with a fibrous surface — comes precisely from its non-parallel fibres.

FEATURE	WORSTED	SEMI-WORSTED	WOOLLEN
Combing	Yes — defining stage	No	No
Gilling / drawing	Extensive, before and after combing	Yes, worsted-type	None; condenser instead
Fibre length used	Long, uniform	Medium to long	Short to medium; noil and blends
Fibre arrangement in yarn	Highly parallel	Fairly parallel	Deliberately disordered
Yarn character	Smooth, strong, clear, finer counts	Bulkier, softer, medium counts	Soft, bulky, lofty, coarser counts
Typical products	Suitings, fine woven fabrics	Knitting yarns, blankets, furnishing	Tweeds, blankets, flannels, carpets
Relative cost	Highest (combing, noil, many passages)	Middle	Lowest for a given raw material

The systems are tools for different products, not grades of quality: worsted preparation for a bulky blanket yarn wastes money, and a woollen yarn cannot be expected to give a clear suiting surface.

5. Modern developments in long-staple spinning

- **High-draft, high-speed gill boxes.** Modern intersecting gills use finer, denser pinning, improved faller drives and higher delivery speeds, with drafts well beyond older practice, so fewer passages achieve the same control. Automatic can changing and sliver stop motions reduce operative intervention and the faults that come with it.
- **Combing developments.** Rectilinear combs have gained in speed, in electronic setting and monitoring of noil extraction, and in quick-change pin systems; the aim is the same noil quality at higher throughput and with less good fibre carried into the noil.
- **Automatic doffing and transport.** Automatic doffing at spinning, and automated movement of cans, bobbins and packages between stages, cut the labour content of a process whose many passages once made it labour-heavy, and reduce handling damage to sliver and roving.
- **Compact spinning in worsted.** Condensing the drafted fibre strand before twist insertion — the compact principle — has been applied to worsted ring spinning, binding edge fibres into the yarn

body: lower hairiness, better fibre utilisation and improved strength, at the cost of the condensing system's energy and maintenance, the same trade-off examined for compact spinning generally.

- **Monitoring and control.** Sliver evenness monitoring at drawing, autolevelling of draft to correct feed variation, and central collection of production and quality data now link the long preparation chain into one supervised process rather than a row of independent machines.

Glossary

Gill box — drafting machine whose pinned faller bars control long fibres through the drafting zone. **Top** — combed, parallel sliver produced by combing. **Noil** — short fibre and impurity removed in combing. **Rectilinear comb** — comb that works on successive held fringes, combing first the fringe and then its tail through a top comb. **Scouring** — washing greasy wool to remove grease, suint and dirt. **Condenser** — woollen-card delivery that divides the web into strips and rubs them into rovings. **Slubbing** — the soft, low-twist roving produced by a condenser. **Semi-worsted** — worsted-type preparation without combing. **Autolevelling** — automatic draft adjustment to correct sliver-weight variation.

Key points

• Long-staple drafting is controlled by pins (gills, combs), because roller settings for long fibres leave too large a floating zone.
• Worsted sequence: scouring → carding → gilling → combing → gilling → drawing → roving → spinning.
• The rectilinear comb removes short fibre as noil; noil % is a cost and quality signal, not a detail.
• Output weight at any gilling or drawing passage = feed weight × doublings / draft.
• Semi-worsted omits combing; woollen omits gilling and combing and condenses a disordered web instead — different yarn character by design.
• Modern developments — high-draft gills, automatic doffing, autolevelling, worsted compact spinning — speed up and supervise the classical logic without replacing it.

Practice

1. A gill box is fed with 6 ends of sliver, each 10 g/m, at a draft of 8. Calculate the output sliver weight.
2. In one week a combing plant processes 400 kg of prepared sliver at a noil extraction of 14%. Calculate the noil produced and the weight of top.
3. A drafting passage takes in 100 m of sliver and delivers 850 m. Calculate the draft.
4. State four differences between the worsted and woollen systems, covering preparation, fibre arrangement and yarn character.

5. Explain why a sudden rise in noil percentage on unchanged raw material should be investigated rather than accepted.

Answer key

1. Output = 10 × 6 / 8 = 60 / 8 = **7.5 g/m**.
2. Noil = 400 × 14/100 = **56 kg**. Top = 400 − 56 = **344 kg**.
3. Draft = delivered length / feed length = 850 / 100 = **8.5**.
4. Any four, for example: worsted is combed, woollen is not; worsted is gilled and drawn, woollen is condensed with no drawing; worsted fibres lie parallel, woollen fibres are deliberately disordered; worsted yarn is smooth, strong and suited to fine counts, woollen yarn is soft, bulky and lofty at coarser counts.
5. Noil is saleable top fibre diverted to a low-value stream. On unchanged fibre, a rise points to a process cause — damaged or worn comb pins, incorrect settings, poor pre-comb gilling, or fibre being broken in carding — each of which also harms top quality. Accepting it pays the cost twice: in lost top and in a poorer yarn.

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